COURSE INFORMATION

School/Faculty:	**Software Engineering/ Computing**		
Program name:	**Bachelor of Computer Science (Graphics and Multimedia Software) with Honours** **Bachelor of Computer Science (Computer Networks and Security) with Honours** **Bachelor of Computer Science (Bioinfomatics) with Honours** **Bachelor of Computer Science (Data Engineering) with Honours**		
Course code:	**SECJ1023**	**Academic Session/Semester:**	**2025/2026-02**
Course name:	**Programming Technique II**	**Pre/co requisite (course name and code, if applicable):**	**Programming Technique I (SECJ1013)**
Credit hours:	**3**		

Course synopsis	This course introduces object-oriented programming (OOP) using C++ and builds upon the foundational programming knowledge acquired in Programming Technique I. It covers key OOP concepts such as abstraction, encapsulation, association, aggregation, composition, inheritance, and polymorphism, as well as class design, object manipulation, UML class diagrams, and selected advanced features including exceptions and templates. The course is delivered through lectures, active learning activities, laboratory exercises, and project-based learning, with emphasis on hands-on programming, systematic problem-solving, and teamwork. By the end of the course, students will be able to analyze problems using object-oriented approaches, develop structured C++ programs based on OOP principles, and collaboratively design and implement a medium-scale application.			
Course coordinator (if applicable)	Ms. Lizawati binti Mi Yusuf			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Ts. Dr. Johanna binti Ahmad			johanna@utm.my
	Dr. Jumail bin Taliba			jumail@utm.my
	Ms. Lizawati binti Mi Yusuf			lizawati@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO **(MQF Cluster Code)	***Taxonomies and ****Graduate attributes	T&L methods	*****Assessment methods
CLO1	Analyse problems systematically using object- oriented approaches.	PLO1 (C1)	C4 (SC1)	Lecture, Active learning, Project-based learning	T1 (15%), F1 (15%), PR1 (10%)
CLO2	Develop programs using object-oriented principles	PLO2 (C2)	C3 (SC2)	Lecture, Active learning, Project-based learning	E (15%), PR2 (10%), T2 (10%), F2 (20%)
CLO3	Work in a team to develop a medium to complex program as a group mini project, using C++ programming language.	PLO8 (C3F)	A3 (CC2)	Project-based learning	PR3 (5%)

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
*Up to 5 CLO
Refer ***Taxonomies of Learning and ****UTM's Graduate Attributes for measurement of outcomes achievement
*****T – Test; PR – Project; F – Final Exam; E – Exercise.

<MQF Cluster Code>
C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C= Communication Skills, C3D = Digital Skills, C3E = Numercy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Enterpreneurial Skills, C5 = Ethics & Professionalism
<Taxonomies>
Cognitive (C) - C1-C6, Psychomotor (P) - P1-P7, Affective (A) - A1-A5
<Graduate Attributes>
SC1, SC2, SC3 = Scholars, SI1, SI2, SI3, SI4 = Social Intelligence, IT1, IT2, IT3 = Innovative Talent, CC1, CC2, CC3, CC4 = Constructive Citizen, ES1, ES2, ES3, ES4, ES5 = Enterprising Skills

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Conducted through in-class activities such as pair programming, group discussion

2	Project-based learning	Conducted through a group project (3 or 4 students per group). The project is divided into several deliverables.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
Team working skills.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1 16/3	**1.0 Introduction** 1.1 Overview of Programming Paradigms 1.2 Revision on Programming Technique I 1.3 Setting Up Programming Environment	CLO1					3					2	5
Week 2 23/3	1.4 Overview of Object-Oriented Programming Principles 1.5 UML Class Diagram ***Group Formation** for the project and discussion of the project idea.*	CLO1					3				1	1	5
Week 3 30/3	**2.0 Introduction to Classes and Objects** 2.1 Defining Classes 2.2 Creating Objects 2.3 Private Members 2.4 Separating Class Specification from Implementation 2.5 Inline Member Functions ***Exersise 1*** ***Project Deliverable 1** (Project Proposal), W3-5*	CLO2	2		1						1	2	6
Week 4 6/4	**3.0 Constructors and Destructors** 3.1 Constructors 3.2 Passing Arguments to Constructors 3.3 Destructors 3.4 Overloading Constructors 3.5 Copy Constructors	CLO1, CLO2	2		1						1	1	5
Week 5 13/4	**4.0 Class and Object Manipulations** 4.1 Friend of Classes 4.2 Pointers to Objects 4.3 Arrays of Objects 4.4 Objects and Functions 4.5 Operator Overloading ***Exercise 2***	CLO2	2		1						1	2	6
Week 6 20/4	**5.0 String Manipulations** 5.1 The string Class 5.2 String Comparisons 5.3 String Operators 5.4 String Member Functions ***Exercise 3*** ***Project Deliverable 2** (Problem Analysis and Design), W6-9*	CLO1, CLO2	2		1						1	2	6
Week 7 27/4	**6.0 Associations, Aggregations and Compositions** 6.1 Introduction to Associations 6.2 Introduction to Aggregations 6.3 Aggregation Implementations ***MidTerm Test (Written)*** *Chap. 1 -Chap. 5* ***Wed 29 April 2026, 8-10 pm***	CLO1	2		1						1	1	5
Week 8	**MID SEMESTER BREAK**												
Week 9 11/5	6.4 Introduction to Compositions 6.5 Composition Implementations ***Exercise 4*** ***MidTerm Test (Practical)*** *Chap. 1 -Chap. 5* ***Wed 13 May 2026, 8-10 pm***	CLO2	2		1						1	3	7
Week 10 18/5	**7.0 Inheritance** 7.1 Introduction to Inheritance 7.2 Protected Members and Class Access 7.3 Constructors and Destructors in Base and Derived Classes ***Project Deliverable 3** (Interim Progress), W10-12*	CLO1	2		1						1	1	5
Week 11 25/5	7.4 Redefining Base Class Functions 7.5 Class Hierarchies 7.6 Multiple Inheritances	CLO2	2		1						1	2	6
Week 12 1/6	**8.0 Polymorphisms** 8.1 Introduction to Polymorphisms 8.2 Polymorphism and Virtual Member Functions	CLO1	2		1						1	1	5
Week 13 8/6	8.3 Abstract Base Classes and Pure Virtual Functions ***Exercise 5*** ***Project Deliverable 4** (Complete Project), W13-15*	CLO2	2		1						1	2	6
Week 14 15/6	**9.0 Overview of Advanced Features** 9.1 Exceptions 9.2 Templates	CLO1	2		1						1		4
Week 15 22/6	9.3 Containers 9.4 Iterators	CLO3	2		1							2	5
												SUB-TOTAL SLT :	**76**

	Continous Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Exercises €	15			3		3
2	Project Design: Proposal and Problem Analysis and Design (PR1)	10			2	4	6
3	Project Implementation: Interim Progress and Final Outcome (PR2)	10			2	4	6
4	Project Teamworking (PR3)	5			1	1	2
5	Test 1 (T1)	15	2		2	2	6
6	Test 2 (T2)	10	2		1	3	6
					SUB-TOTAL SLT :		**29**

	Summative Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam - Theory (F1)	15	2		1	3	6
2	Final Exam - Practical (F2)	20	3		2	4	9
					SUB-TOTAL SLT :		**15**
					SLT for Assessment:		**44**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	37.50
B	% SLT for Online & Independent Learning Component :	62.50
C	%SLT for Online Component:	26.67
D	% SLT for All Practical Component:	10.00
D1	% SLT for F2F Physical Practical Component:	10.00
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

Computer Lab (with internet connections): For Lab Exercises and practical tests
Software: C++ IDE such as Microsoft Visual Studio Code, Dev C++, etc.

References (include required and further readings, and should be the most current)

Main references:
Gaddis, T. (2018). Starting out with C++: From control structures through objects (9th ed.). Pearson.
Other references:
1. Deitel, P., & Deitel, H. (2020). C++ how to program (10th ed.). Pearson.
2. Gaddis, T. (2020). Starting out with C++: Early objects (10th ed.). Pearson.
3. Savitch, W. (2019). Problem solving with C++ (10th ed.). Pearson.
4. Malik, D. S. (2021). C++ programming: From problem analysis to program design (9th ed.). Cengage Learning.

Other additional information (if applicable)

Academic honesty and plagiarism:

Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

Disclaimer:

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ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:		**Certified by:**	
Name:	Lizawati binti Mi Yusuf	**Name:**	Prof. Madya Dr. Adham bin Isa
Signature:		**Signature:**	
Date:	9/3/2026	**Date:**	9/3/2026